Hyunwoo Park

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EDUCATION

Yonsei University Seoul, Korea
B.S. / Mechanical Engineering Mar.2016 - Feb.2022*
Cumulative GPA: 3.60 / 4.00
Class Rank: 37 / 142

EXPERIENCES

StradVision Seoul, Korea
Motion Planning & Vehicle Control Team, Algorithm Engineer Jan.2026 - Present

Projects

Development of Motion Planning for Autonomous Parking Vehicle (Jan.2026 - Present)

ThorDrive Seoul, Korea
Motion Planning Engineer Jan.2022 - Jan.2026

Projects

Development of Open Space Planner (Jan.2022 - Aug.2022)
- Developed path planner for autonomous vehicles in open spaces, such as parking lots.
- Hybrid A*, and sequential quadratic programming(a numerical optimization technique) were used.

Development of Motion Planner for Public Roads and Airport Apron (Aug.2022 - Apr.2025)
- Developed motion planning features such as avoiding static obstacles, overtaking parked cars, emergent collision avoidance, and handling general road scenarios.

Development of Driving Strategy in Occluded Areas (Dec.2022 - Jun.2023)
- Developed velocity planning technique for navigating through occluded areas while assessing occlusion risks.
- Published in IEEE Robotics and Automation Letters (RA-L), 2023

Development of B-Spline-Based Trajectory Optimization for Autonomous Vehicles (Dec.2022 - Dec.2024)
- Developed a B-spline-based trajectory optimization method using incremental path flattening to generate smooth, collision-free, and kinematically feasible trajectories.
- Published in IEEE Transactions on Intelligent Transportation Systems (T-ITS), 2024.

Development of Reinforcement Learning-Based Trajectory Planner (Oct.2023 - Apr.2024)
- Developed a trajectory planning technique using deep reinforcement learning.

Development of Sim-to-Real Transfer Method for Reinforcement Learning Agents (Jul.2024 - Mar.2026)
- Developing a sim-to-real transfer method that combines adversarial domain adaptation with a learned world model to generate imagined trajectories for reinforcement learning.
- Published in European Conference on Computer Vision (ECCV), 2026

* Including military service (Apr.2018 - Dec.2019)

Development of a Dynamic Obstacle Avoidance Module for Autonomous Mobile Robots in Logistics Factories (Apr.2025 - Jan.2026)

- Developing a dynamic obstacle avoidnace module based on the NAV2 framework.

Programmers Co. Seoul, Korea
Autonomous Driving Engineer Course Mar.2021 - Sep.2021

Projects

Autonomous Driving Team Projects (Mar.2021 - Sep.2021)

- Learned how the autonomous vehicle works and developed software to execute various tasks, including static obstacle avoidance, parking, and lane following, using a 1/10 scale model car equipped with lidar and camera sensors.

Yonsei University Seoul, Korea
Multi-Disciplinary, Multi-Physics, Multi-Scale Design and Optimization Lab Internship Aug.2020 - Dec.2020

Projects

Off-The-Ground Mobility, 2020 Alchemist project, Ministry of trade, Industry and Energy, Korea (Aug.2020 - Dec.2020)

- Designed a new concept of mobility that allows a person to board and control while floating above the ground surface. Designed an initial model and verified it using MATLAB Simulink.

PUBLICATIONS

- Park, H., 2024. Trajectory Planning for Autonomous Vehicle Using Iterative Reward Prediction in Reinforcement Learning. arXiv preprint arXiv:2404.12079.
- Choi, J., Chin, H., Park, H., Kwon, D., Lee, S. and Baek, D., 2024. Safe and Efficient Trajectory Optimization for Autonomous Vehicles using B-spline with Incremental Path Flattening. IEEE Transactions on Intelligent Transportation Systems (T-ITS)
- Park, H., Choi, J., Chin, H., Lee, S.H. and Baek, D., 2023. Occlusion-aware risk assessment and driving strategy for autonomous vehicles using simplified reachability quantification. IEEE Robotics and Automation Letters. (RA-L)
- Park, H., and Lee, S.H., 2026. Domain Adaptation with Adaptive Imagination for Visual Reinforcement Learning under Limited Target Data. European Conference on Computer Vision (ECCV)

INVITED TALKS & LECTURES

Invited Speaker, Department of Mobility Engineering, Ajou University, (Apr.2026)
Delivered a one-hour industry seminar on autonomous driving engineering and motion planning.

SKILL & KNOWLEDGE SET

Skill Set: C++, Python, ROS
Knowledge Set: Motion Planning, Optimization, Reinforcement Learning, Domain Adaptation

HONORS AND AWARDS

- Employee of the year, ThorDrive, 2023
- Yonsei University, 1st semester, 2016 - HONORS
- Yonsei University, 1st semester, 2020 - HONORS

CERTIFICATION

Udacity Mentor Certification (Oct.2022 - Present)

- Motion Planning and Decision Making for Autonmous Vehicles

TOEIC (975, ENGLISH) (Jan.2021)
OPIc (AL , ENGLISH) (Feb.2021)